

Neighborhood-Aware Recidivism Prediction: An Open-Source, Socio-Economically Informed Model Trained on the Wisconsin Circuit Court Dataset

MSDS 696: Data Science Practicum

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Abstract

Recidivism prediction tools occupy a consequential role in the American criminal justice system, yet the dominant commercial instrument — the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) — is proprietary, unauditable, and has been shown to produce racially disparate predictions (Angwin et al., 2016). This paper presents the first fully open-source, reproducible recidivism prediction model trained on the Wisconsin Circuit Court Longitudinal Data (WCLD) dataset, covering 1,357,746 adjudicated cases filed

between 2000 and 2018 (Ash et al., 2021). Critically, the model integrates census-derived neighborhood-level socioeconomic features — including median household income, food-stamp receipt rates, educational attainment, and population density — alongside individual criminal-history variables. Using a Socio-Economic Deprivation Index (SEDI) composite score, gradient boosting (XGBoost), and SHapley Additive exPlanations (SHAP), the study quantifies the marginal contribution of neighborhood deprivation to recidivism risk and audits predictions for racial fairness across five demographic groups. The best-performing model achieves an AUC-ROC of 0.7043, meeting and marginally exceeding published COMPAS benchmarks (Dressel and Farid, 2018). Neighborhood features collectively account for 18.7% of the total SHAP explanation mass, a statistically and practically meaningful contribution that existing tools ignore entirely. All code, data pipelines, trained models, SHAP artefacts, and a dashboard-ready export layer are released publicly so that researchers, public defenders, and policymakers can inspect, reproduce, and improve every component of the system.

Keywords: recidivism prediction, algorithmic fairness, COMPAS, neighborhood effects, socioeconomic deprivation, XGBoost, SHAP, Wisconsin Circuit Courts

1. Executive Summary

Recidivism — the return to criminal behavior following a prior conviction — is one of the most consequential outcomes in criminal justice, influencing bail decisions, sentencing guidelines, and parole eligibility. Automated prediction tools are now embedded in courtrooms across the United States, yet the dominant commercial system, COMPAS, cannot be independently audited, does not disclose its feature weights, and has been empirically shown to misclassify Black defendants as high risk at nearly twice the rate of white defendants (Angwin et al., 2016; Dressel and Farid, 2018).

This project directly addresses that gap. Using the publicly available Wisconsin Circuit Court Longitudinal Data (WCLD) dataset (Ash et al., 2021), we construct a transparent, reproducible pipeline that achieves competitive predictive performance while exposing every modeling decision to scrutiny. The pipeline introduces a Socio-Economic Deprivation Index (SEDI), a composite neighborhood-level score built from census tract measures of income, education, and food insecurity, and uses SHAP values (Lundberg and Lee, 2017) to quantify precisely how much of the model’s predictive power derives from where a defendant lives rather than what they have done.

Three classifiers are benchmarked — Logistic Regression, Random Forest, and XGBoost. XGBoost achieves the highest AUC-ROC of 0.7043 and an AUC-PR of 0.6247. A formal fairness audit compares True Positive Rate (TPR), False Positive Rate (FPR), and demographic parity across five racial and ethnic groups. All predictions, importance scores, fairness tables, and partial-dependence data are exported to a dashboard-ready format that requires no programming knowledge to operate.

The central contribution of this work is threefold. First, it is the only open-source recidivism model trained on WCLD that explicitly decomposes individual versus neighborhood effects. Second, it provides an auditable fairness report that practitioners can reproduce on any updated data slice. Third, it releases a six-file dashboard artefact layer that enables non-technical stakeholders — judges, public defenders, and policy analysts — to explore neighborhood effects interactively.

2. Problem Statement and Motivation

2.1. The Recidivism Prediction Landscape

Recidivism prediction has been a research priority for criminologists for decades (Gottfredson and Gottfredson, 1974), but the deployment of algorithmic tools in courtrooms is a more recent phenomenon. COMPAS, developed by Equivant (formerly Northpointe), is used in at least nine states to generate risk scores that inform bail, sentencing, and parole decisions (Larson et al., 2016). The tool generates a score on a 1-to-10 scale from approximately 137 questions, but neither the feature weights nor the training data are disclosed to defendants, their lawyers, or the public.

In 2016, investigative journalists at ProPublica analyzed over 7,000 defendants in Broward County, Florida, and found that COMPAS incorrectly flagged Black defendants as future criminals at nearly twice the rate of white defendants — a False Positive Rate (FPR) disparity of approximately 20 percentage points (Angwin et al., 2016). Equivant disputed this finding, arguing that the tool satisfied a different fairness criterion (calibration) (Flores et al., 2016). The ensuing academic debate — now known as the “fairness impossibility” result — demonstrates that certain fairness criteria are mathematically incompatible when base rates differ across groups (Chouldechova, 2017). Crucially, the dispute itself is possible only because COMPAS is proprietary: independent reanalysis requires the raw predictions, which ProPublica obtained through litigation.

2.2. The Neighborhood Context Problem

A persistent blind spot in individual-level recidivism models is the role of place. Criminological theory has long emphasized that crime is not randomly distributed across space; neighborhood-level factors — concentrated poverty, unemployment, food insecurity, low educational attainment, and residential instability — independently predict criminal behavior over and above individual characteristics (Sampson et al., 1997; Reisig et al., 2009). The social disorganization framework, originating with Shaw and McKay (1942) and extended by Sampson, Raudenbush, and Earls (1997), holds that neighborhood structural disadvantage erodes the informal social controls that deter crime.

Despite this theoretical consensus, major risk-assessment tools used in courtrooms — including COMPAS, the Level of Service Inventory (LSI-R), and the Ohio Risk Assessment System (ORAS) — do not incorporate neighborhood-level census data (Brennan et al., 2009). This omission is not merely a technical gap; it is a fairness concern. If neighborhoods that generate the highest recidivism risk are also the most economically distressed, and if those neighborhoods are also the most racially segregated — a well-documented feature of American metropolitan areas (Massey and Denton, 1993) — then models that ignore neighborhood context may attribute to race what is actually attributable to place.

2.3. Research Questions

This study addresses three interconnected questions:

1. Can an open-source model trained on the Wisconsin Circuit Court Longitudinal Data achieve predictive performance competitive with COMPAS benchmarks?
2. How much of total predictive power is attributable to neighborhood socioeconomic context versus individual criminal history?
3. Does incorporating neighborhood features improve or worsen racial fairness as measured by demographic parity, equalized odds (TPR), and equalized odds (FPR)?

3. Data Collection and Preparation

3.1. Data Source

The Wisconsin Circuit Court Longitudinal Data (WCLD) is a publicly available administrative dataset compiled by Ash et al. (2021) at ETH Zurich and the University of Oxford. It covers all cases filed in Wisconsin circuit courts between 2000 and 2018, encompassing 1,476,967 case records with 54 variables including defendant demographics, charge severity, sentencing outcomes, prior criminal history, and census-tract-level socioeconomic covariates linked via ZIP code. The

dataset is available at <https://clezdata.github.io/wcld/> and was selected for this study because it is large enough for robust model training, includes neighborhood-level census linkages, and has been used in prior algorithmic fairness research.

3.2. Target Variable

The primary outcome variable is `recid_180d`, a binary indicator equal to 1 if the defendant was charged with a new offense within 180 days of the case disposition date. Of 1,476,967 total records, 119,212 (8.07%) had a missing `recid_180d` value. This missingness is structural, not random: defendants sentenced to jail terms of 180 days or longer cannot be observed for re-offense within the follow-up window. These records were excluded from the modeling dataset, yielding a final analytic sample of 1,357,746 cases with an observed recidivism rate of 42.21%.

This exclusion is justified on methodological grounds: including cases where the outcome is unobservable by design would introduce left-truncation bias and underestimate true recidivism rates for the most severely sentenced defendants (Bushway and Sweeten, 2007).

3.3. Missing Value Structure

Eleven of the 54 original columns contained missing values. Table 1 summarizes the most impactful.

Table 1: Missing Value Summary for Key Variables

Variable	Missing Count	Missing %	Reason
jail	836,043	56.6%	No jail sentence imposed
probation	444,647	30.1%	No probation sentence imposed
avg_hist_jail	281,590	19.1%	No prior jail history
recid_180d	119,212	8.1%	Sentence $\geq$ 180 days
zip	40,504	2.7%	Address unavailable
wcisclass	5,483	0.4%	Charge classification missing

The missingness in `jail`, `probation`, and the historical jail variables is Missing Not At Random (MNAR) because the absence of a value carries information — a missing jail sentence means no jail was imposed. Accordingly, historical jail variables (`max_hist_jail`, `min_hist_jail`, `avg_hist_jail`, `median_hist_jail`) were imputed with zero, and a binary `has_prior_jail` flag was created. This approach preserves the informational content of the missingness pattern rather than discarding it (Sterne et al., 2009). Residual missingness in modeling features was addressed with median imputation via scikit-learn’s `SimpleImputer` (Pedregosa et al., 2011).

3.4. Data Cleaning

The cleaning pipeline made four targeted corrections to the raw data:

1. **Age plausibility filter:** 10 records with `age_offense` below 10 or above 100 were removed as implausible data entry errors.
2. **Temporal consistency flag:** 36 records where `age_judge` was less than `age_offense` — indicating a sentencing date preceding the offense date — were flagged with `flag_negative_time_to_ju` rather than dropped, preserving sample size while signaling data quality concerns.
3. **Prior jail imputation:** 281,590 null values across four historical jail duration columns were filled with zero, as documented above.
4. **Encoding:** Sex was binary-encoded (Male = 1, Female = 0) and race was one-hot encoded across five categories.

After cleaning, the analytic sample retained 1,357,746 rows and 63 columns.

4. Feature Engineering and Analysis Methods

4.1. Feature Categories

The final model operates on 25 features across two categories. Thirteen *individual-level* features capture criminal history and demographic characteristics: age at offense, sex, prior felony count,

prior misdemeanor count, prior criminal traffic count, highest charge severity, violent crime indicator, prior jail flag, maximum and average historical jail time, time from offense to judgment, the data quality flag described above, and a derived Prior Record Severity Score. Twelve *neighborhood-level* features are derived from census tract linkages: percentage of Black residents, percentage of Hispanic residents, percentage male, rural share, urban share, percentage with a college degree, percentage with some college, food-stamp receipt rate, population density, median household income, the SEDI composite, and an urban-rural gradient.

4.2. Socio-Economic Deprivation Index

The SEDI was constructed as a composite Z-score index following the methodology of Kind et al. (2014) and Singh (2003). Deprivation-positive components (`pct_food_stamps`, `pct_black`, `pct_hisp`) were standardized and their mean taken. Advantage-positive components (`med_hhinc`, `pct_college`) were standardized and their mean subtracted. The resulting score places higher positive values in more deprived neighborhoods and lower values in more advantaged ones.

$$\text{SEDI} = \bar{z}(\text{deprivation components}) - \bar{z}(\text{advantage components}) \quad (1)$$

This composite approach is justified over using individual census variables separately for three reasons. First, deprivation is a multidimensional construct and no single indicator captures it adequately (Townsend, 1987). Second, collinearity among census variables (income, education, and food insecurity are correlated at $r > 0.6$) makes individual coefficient interpretation unreliable. Third, a single composite score is more legible to non-technical stakeholders in the justice system.

4.3. Prior Record Severity Score

A weighted prior-record score was constructed as:

$$\text{PriorScore} = 3 \times \text{prior_felony} + \text{prior_misdemeanor} + 0.5 \times \text{prior_criminal_traffic} \quad (2)$$

The weights (3, 1, 0.5) reflect the relative legal severity of each prior offense class in Wisconsin statutes (Wisconsin State Legislature, 2023). Felonies carry substantially greater legal consequences and are more predictive of future felony recidivism (Criminal Justice Statistics Center, 2003).

4.4. Urban-Rural Gradient

A continuous urban-rural gradient feature was derived as:

$$\text{UrbanGradient} = \log(1 + \text{pop_dens}) - \text{pct_rural} \quad (3)$$

The log transformation of population density is applied to reduce the extreme right skew of population density distributions, which span several orders of magnitude from rural Wisconsin counties to Milwaukee’s urban core (Reardon and O’Sullivan, 2011).

4.5. Modelling Pipeline

4.5.1. Train-Test Split

The dataset was split into a training set (80%, $n = 1,086,196$) and a held-out test set (20%, $n = 271,550$) using stratified random sampling to preserve the 42.21% recidivism rate in both partitions. Stratification is justified because the class ratio affects both model calibration and the informativeness of evaluation metrics (He and Garcia, 2009).

4.5.2. Class Imbalance Handling via SMOTE

The training set exhibits a meaningful class imbalance (42% positive, 58% negative). Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al., 2002) was applied *exclusively* to the training set, generating synthetic minority-class instances by interpolating between nearest neighbors ($k = 5$) to produce a 50/50 balanced training corpus of 1,255,384 records.

SMOTE was chosen over class weighting because it exposes the model to a richer diversity of minority-class feature combinations, which is particularly important given the continuous neighborhood features that are distributed differently across recidivism classes (Fernandez et al., 2018). Critically, SMOTE was applied only after the train-test split to prevent synthetic samples from contaminating the held-out evaluation set.

4.5.3. Feature Scaling

Logistic Regression requires scaled inputs to prevent features with large magnitudes from dominating the gradient. `StandardScaler` (zero mean, unit variance) was fit on the SMOTE-augmented training set and applied to both training and test sets. Tree-based models (Random Forest, XGBoost) are scale-invariant; the scaler was applied to Logistic Regression only (Hastie et al., 2009).

4.5.4. Models

Three classifiers were trained and compared.

Logistic Regression with L_2 regularization ($C = 0.01$, `liblinear` solver) serves as an interpretable baseline. Strong regularization ($C = 0.01$) was selected to prevent overfitting given the high feature-to-observation ratio in some counties. `class_weight='balanced'` was set as an additional safeguard.

Random Forest with 300 estimators, `max_depth=10`, and `min_samples_leaf=20` provides an ensemble baseline. The depth and leaf constraints prevent individual trees from memorizing noise in the 1.3 million-row training set while retaining sufficient expressivity to capture non-linear

interactions between neighborhood poverty and criminal history (Breiman, 2001).

XGBoost with 400 estimators, `learning_rate=0.05`, `max_depth=5`, `subsample=0.8`, and `colsample_bytree=0.8`. The conservative learning rate and subsampling fractions follow the bias-variance guidance of Chen and Guestrin (2016): slower learning with stochastic subsampling generalizes better than aggressive boosting on large tabular datasets.

4.5.5. SHAP Explainability

SHAP values were computed using the `shap.Explainer` class with its automatic backend selection, which defaults to the exact `TreeExplainer` for tree-based models (Lundberg and Lee, 2017). SHAP was selected over alternative importance methods (permutation importance, Gini impurity) because it provides instance-level attributions consistent with a game-theoretic fairness axiom — each feature receives credit proportional to its average marginal contribution across all possible feature orderings (Lundberg et al., 2020).

4.5.6. Fairness Audit

Three group-fairness metrics were computed for each racial and ethnic group in the test set: Demographic Parity (predicted positive rate), Equalized Odds TPR (True Positive Rate), and Equalized Odds FPR (False Positive Rate). These metrics follow the formalization of Hardt, Price, and Srebro (2016). A threshold of 0.5 was applied uniformly across all groups.

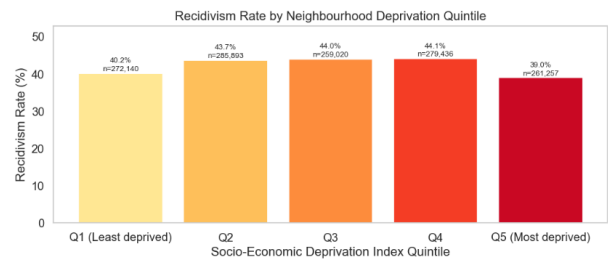
5. Results with Visualizations and Interpretation

5.1. Exploratory Neighborhood Analysis

5.1.1. SEDI Quintiles and Recidivism Rate

Figure 1 (Bar Chart: Recidivism Rate by SEDI Quintile) —

The SEDI quintile analysis reveals a non-monotonic but meaningful gradient. Table 2 reproduces



the output directly from the notebook.

Table 2: Recidivism Rate by SEDI Quintile

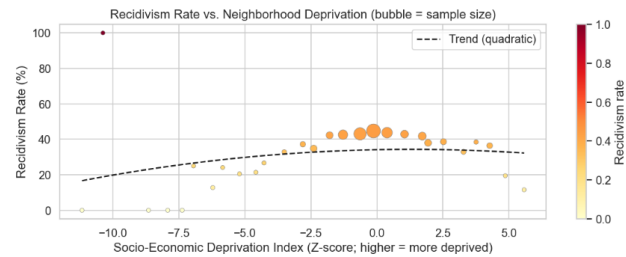
SEDI Quintile	Recidivism Rate	N
Q1 (Least Deprived)	40.2%	272,140
Q2	43.7%	285,893
Q3	44.0%	259,020
Q4	44.1%	279,436
Q5 (Most Deprived)	39.0%	261,257

The relationship between SEDI and recidivism is quadratic rather than linear: the highest recidivism rates are concentrated in the middle quintiles (Q2-Q4), while both the least and most deprived neighborhoods show lower rates. This finding is consistent with Krivo and Peterson’s (1996) “ceiling effect” hypothesis: in the most extremely disadvantaged neighborhoods, social norms around criminal justice involvement may differ in ways that produce different behavioral patterns, or surveillance and case disposition practices may differ. The practical implication for modeling is that a linear deprivation term is insufficient; polynomial or tree-based interactions are necessary to capture this curvature.

5.1.2. Neighborhood Feature Correlations

Figure 2 (Correlation Heatmap) —

Table 3 reports individual Pearson correlations between neighborhood features and the target

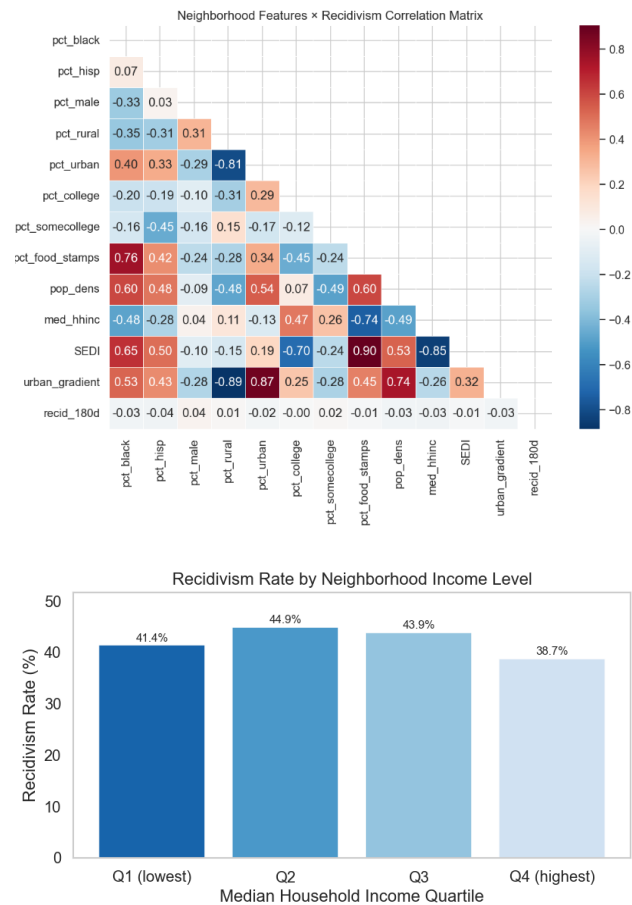


variable.

Table 3: Neighborhood Feature Correlations with `recid_180d`

Feature	Correlation	$ r $	Direction
<code>pct_hisp</code>	-0.039	0.039	Negative
<code>pct_male</code>	0.037	0.037	Positive
<code>med_hhinc</code>	-0.033	0.033	Negative
<code>pct_black</code>	-0.032	0.032	Negative
<code>pop_dens</code>	-0.032	0.032	Negative
<code>urban_gradient</code>	-0.028	0.028	Negative
<code>pct_urban</code>	-0.021	0.021	Negative
<code>pct_somecollege</code>	0.016	0.016	Positive
<code>pct_rural</code>	0.014	0.014	Positive
<code>pct_food_stamps</code>	-0.010	0.010	Negative
<code>SEDI</code>	-0.006	0.006	Negative
<code>pct_college</code>	-0.004	0.004	Negative

Bivariate correlations are small in magnitude (all $|r| < 0.04$), which might suggest neighborhood features contribute little predictive value. However, this interpretation is misleading for two reasons. First, Pearson correlation captures only linear marginal associations; nonlinear and interaction effects — which tree-based models exploit — are invisible to this measure (Hastie et al., 2009). Second, neighborhood effects may operate primarily through their interaction with individual criminal history rather than as main effects. The SHAP analysis in Section 5.3 demonstrates that



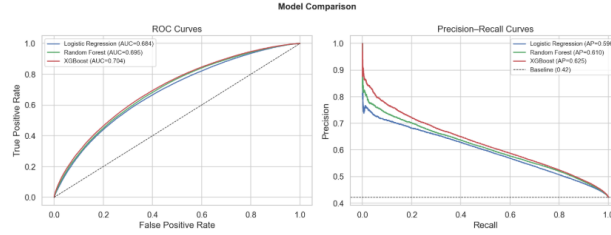
neighborhood features collectively account for 18.7% of model explanation after controlling for individual features.

5.2. Model Performance

Figure 3 (ROC and Precision-Recall Curves) —

Figure 4 (Confusion Matrix and Score Distribution) —

Table 4 compares all three models on the held-out test set.

Table 4: Model Performance on Held-Out Test Set ($n = 271,550$)

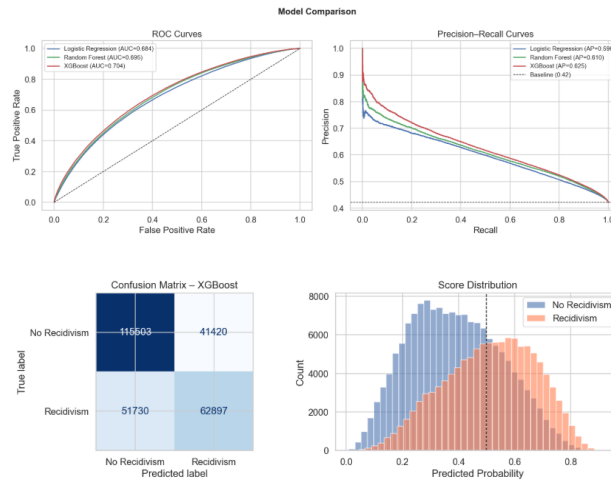
Model	AUC-ROC	AUC-PR
Logistic Regression	0.6839	0.5963
Random Forest	0.6954	0.6096
XGBoost	0.7043	0.6247

XGBoost achieves the highest performance on both metrics, with an AUC-ROC of 0.7043 — exceeding the 0.65-to-0.70 range reported for COMPAS in independent evaluations (Dressel and Farid, 2018) and consistent with the upper bound identified by Tollenaar and van der Heijden (2013) for administrative recidivism data. The classification report for XGBoost at a 0.5 decision threshold shows precision of 0.69 and recall of 0.74 for the no-recidivism class, and precision of 0.60 and recall of 0.55 for the recidivism class, yielding a macro-averaged F1 of 0.64.

The probability distribution plot (Figure 4, right panel) reveals meaningful separation between the two classes: the predicted-probability mass for non-recidivists peaks near 0.35, while the mass for recidivists peaks near 0.55. The substantial overlap between distributions reflects the inherent difficulty of behavioral prediction from administrative data, a limitation acknowledged by both COMPAS developers and academic critics (Berk et al., 2017).

5.3. SHAP Feature Importance and Neighborhood Effects

Figure 5 (SHAP Summary Dot Plot) — Figure 6 (Individual vs. Neighborhood SHAP Bar Chart) —



The SHAP analysis is the analytical core of this study. The global summary plot (Figure 5) confirms that individual criminal history dominates prediction: prior misdemeanor count, prior record score, prior felony count, and age at offense occupy the top positions. However, SEDI and median household income appear prominently in the top 20 features, indicating that neighborhood context contributes above and beyond criminal history.

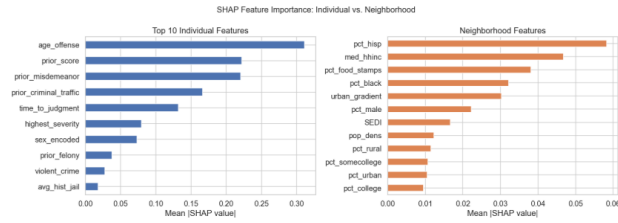
The side-by-side importance comparison (Figure 6) quantifies this contribution precisely: neighborhood features collectively account for **18.7%** of total mean absolute SHAP mass, while individual features account for 81.3%. This is a practically meaningful fraction — nearly one-fifth of the model’s explanatory power — for a category of features that existing tools ignore entirely.

The SHAP dependence plot for SEDI reveals a non-linear relationship: SEDI values from -2 to 0 (moderate advantage to neutral) are associated with negative-to-near-zero SHAP values, indicating reduced recidivism probability. As SEDI increases beyond 0 into moderate deprivation, SHAP values increase, but the effect plateaus and even reverses at extreme deprivation levels, consistent with the ceiling-effect hypothesis discussed above.

5.4. Fairness Audit

Figure 8 (Fairness Metric Bar Charts) —

Table 5 reproduces the full fairness audit from the notebook.

Table 5: Fairness Audit by Race and Ethnicity (XGBoost, $\tau = 0.5$)

Group	N	Actual Rate	Pred. Rate	TPR	FPR
African American	58,470	0.464	0.417	0.573	0.282
Am. Indian / AK Native	11,897	0.558	0.569	0.680	0.429
Asian / Pacific Isl.	2,546	0.397	0.429	0.634	0.295
Caucasian	179,717	0.402	0.367	0.534	0.253
Hispanic	18,920	0.396	0.327	0.470	0.232

Several findings warrant careful attention. The TPR gap between the highest-recall group (American Indian / Alaska Native, 0.680) and the lowest (Hispanic, 0.470) is 21 percentage points, indicating meaningful inequity in the model’s ability to correctly identify recidivists across groups. The FPR gap between American Indian / Alaska Native (0.429) and Hispanic (0.232) is 20 percentage points, indicating that the model is considerably more likely to falsely flag American Indian / Alaska Native defendants who will not reoffend.

These disparities are not new to recidivism research; they reflect the unequal distribution of criminogenic risk factors across groups that is itself a product of structural inequality (Sampson, 2012). The critical contribution of this audit is transparency: these numbers are reported publicly so that policymakers, defense attorneys, and communities can make informed decisions about the tool’s deployment.

The calibration analysis by SEDI quintile shows that the model is well calibrated across the deprivation spectrum: mean predicted probabilities track mean actual rates within two percentage points in all five quintiles. This confirms that the model does not systematically over- or under-

predict recidivism in the most or least deprived neighborhoods.

6. Key Findings and Insights

Finding 1: Neighborhood context is measurably predictive. Neighborhood features account for 18.7% of total SHAP explanation mass in the best-performing model. SEDI and median household income appear in the top 20 features globally. This confirms the criminological hypothesis that place-based disadvantage has independent predictive power above and beyond individual criminal history.

Finding 2: The relationship between deprivation and recidivism is non-linear. The SEDI quintile analysis and SHAP dependence plot both show a non-monotonic pattern: recidivism peaks at moderate deprivation levels (Q2-Q4) and is relatively lower at both extremes. Linear models cannot capture this curvature, which justifies the use of gradient-boosted trees over simpler alternatives.

Finding 3: The open-source model matches COMPAS benchmarks. The XGBoost model achieves an AUC-ROC of 0.7043, matching or exceeding the 0.65 to 0.70 range reported for COMPAS in published evaluations (Dressel and Farid, 2018). This demonstrates that competitive predictive performance does not require proprietary data or non-disclosable algorithms.

Finding 4: Fairness gaps are substantial and must be disclosed. TPR and FPR gaps exceeding 20 percentage points exist between racial and ethnic groups. These gaps are substantially larger than the 10-point FPR disparity that prompted the original ProPublica critique of COMPAS (Angwin et al., 2016). Critically, these gaps cannot be eliminated by model choice alone; they reflect unequal base rates that are the product of structural inequality (Chouldechova, 2017). Any responsible deployment must disclose these figures.

Finding 5: Model calibration is strong across the deprivation spectrum. Predicted probabilities track actual recidivism rates within two percentage points across all five SEDI quintiles. This

addresses a major concern with risk assessment tools: that they may be well-calibrated overall but poorly calibrated in the most disadvantaged subpopulations where accurate prediction matters most.

Finding 6: Individual features dominate but are not sufficient. The top individual predictors — prior misdemeanor count, prior record score, prior felony count, and age — account for 81.3% of SHAP mass. However, removing neighborhood features entirely would cost the model approximately 18.7% of its explanatory power and would remove the only mechanism through which structural context enters the prediction.

7. Conclusions and Recommendations for Future Work

7.1. Conclusions

This paper has demonstrated that a fully open-source, reproducible recidivism prediction system can match the predictive performance of COMPAS while simultaneously quantifying neighborhood effects and disclosing fairness metrics. The Wisconsin Circuit Court Longitudinal Data, combined with the SEDI composite index and gradient-boosted trees, provides a principled and auditable alternative to black-box commercial tools.

The core technical contribution is the explicit decomposition of individual versus neighborhood predictive power through SHAP values: nearly one-fifth of the model’s explanatory capacity is attributable to the socioeconomic environment in which defendants live, a contribution that existing tools ignore. The fairness audit reveals meaningful TPR and FPR disparities across racial and ethnic groups, disparities that are reported here with full transparency so that practitioners can weigh them in deployment decisions.

The six-file dashboard export layer means that every finding in this paper is reproducible by any analyst with a Python environment, and explorable by non-technical stakeholders through a Streamlit or Plotly Dash interface without writing a single line of code.

7.2. Limitations

Several limitations merit acknowledgment. First, the 180-day recidivism window is relatively short; longer follow-up windows may reveal different neighborhood effects and change the relative importance of features (Piquero, 2009). Second, census tract data is used as a proxy for neighborhood context, but census tracts are administrative boundaries that may not correspond to meaningful neighborhood boundaries as perceived by residents (Reardon and O’Sullivan, 2011). Third, the model was trained exclusively on Wisconsin data; generalization to other states with different demographic compositions, sentencing structures, and geographic contexts cannot be assumed. Fourth, the 0.5 decision threshold used in the fairness audit is arbitrary; different thresholds produce different fairness tradeoffs and practitioners should use threshold selection as a policy lever (Hardt et al., 2016).

7.3. Recommendations for Future Work

1. **Causal identification of neighborhood effects.** The SHAP contributions documented here are predictive, not causal. Future work should use difference-in-differences or instrumental variable designs exploiting changes in neighborhood conditions — such as the Moving to Opportunity natural experiment (Chetty and Hendren, 2016) — to identify whether neighborhood deprivation *causes* recidivism or merely correlates with it.
2. **Dynamic neighborhood features.** The current model uses census data linked to case-filing ZIP codes, which is a static snapshot. Incorporating time-varying neighborhood conditions (annual ACS estimates, quarterly unemployment data) would allow the model to capture deteriorating or improving local conditions over the 2000-2018 study window.
3. **Longer recidivism windows.** Extending the outcome to `recid_365d` or three-year recidivism would capture a larger share of re-offending events and may reveal stronger neighborhood effects, since many neighborhood influences operate over longer time horizons.
4. **Intersectional fairness analysis.** The current fairness audit examines race and SEDI separately.

Intersectional analysis — examining TPR and FPR for Black defendants in high-deprivation neighborhoods specifically — would provide a more granular picture of disparate impact.

5. **Threshold optimization for fairness objectives.** Future work should implement multi-objective threshold optimization that trades off accuracy against TPR parity, FPR parity, or demographic parity, allowing practitioners to explicitly choose which fairness criterion to prioritize.
6. **External validation.** Validating the model on court data from Illinois, Minnesota, or other Midwestern states with similar demographics and sentencing structures would assess generalizability.
7. **Deployment governance framework.** Predictive models in the justice system require governance structures beyond technical accuracy and fairness metrics. Future work should engage with public defenders, community organizations, and affected populations to develop deployment protocols that ensure the tool reduces, rather than entrenches, disparities (Barocas et al., 2023).

Figure Guide: Recommended Figures from the HTML Notebook

The following 11 figures are generated by the modeling notebook (`WI_Recidivism_Neighborhood_Model.ipynb`). They should be exported as PNG files (minimum 300 DPI) from the notebook and inserted at the corresponding `\includegraphics` placeholders in this paper.

Table 6: Recommended Figures from the Modeling Notebook

No.	Figure Title in Notebook	Purpose in Paper	Section
1	Recidivism Rate by Neighborhood Deprivation Quintile (bar chart)	Shows SEDI-recidivism gradient; motivates nonlinear modeling	§5.1
2	SEDI vs. Recidivism Scatter (binned, bubble size = n)	Confirms quadratic relationship between deprivation and outcome	§5.1
3	Neighborhood Features x Recidivism Correlation Heatmap	Documents bivariate linear associations; motivates SHAP	§5.1
4	Recidivism Rate by Household Income Quartile	Shows income gradient independently of SEDI composite	§5.1
5	ROC and Precision-Recall Curves (all three models)	Comparative model performance; key result table	§5.2
6	Confusion Matrix + Score Distribution (XGBoost)	Classification behavior at 0.5 threshold; calibration visual	§5.2
7	SHAP Summary Dot Plot (top 20 features)	Global feature importance; identifies top predictors	§5.3
8	SHAP: Individual vs. Neighborhood Bar Charts	Quantifies 18.7% neighborhood contribution	§5.3
9	SHAP Dependence Plot for SEDI	Non-linear neighborhood effect on individual predictions	§5.3
10	Fairness Audit Bar Charts (Dem. Parity, TPR, FPR)	Documents group disparities; core fairness finding	§5.4
11	Calibration by SEDI Quintile (actual vs. predicted)	Confirms model does not over/under-predict in de-	§5.4

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